Math 252
Spring 2019

Name ______________________ Section: A B

Test 2

Instructions

1. As you do the problems on this test, be sure to show your work.
2. If you use your calculator, enter both what you typed into your calculator and the result.
3. If you round, round late and show the value (3 or 4 significant digits) both before and after rounding.

1 Suppose $p(A) = 0.6$, $p(B) = 0.3$, and $p(A \cap B) = 0.2$.

a) What is $p(A \cup B)$?

b) What is $p(A \mid B)$?

c) What is $p(B \mid A)$?

d) Are events A and B **independent**? Explain how you know.

e) Are events A and B **mutually exclusive**? Explain how you know.

2 Terry rolls 13 standard (six-sided) dice. What is the probability that Terry rolls **exactly 3** 6's?

3 Each of these problems deals with probabilities when **four 4-sided dice** are rolled.

a) How many dice are being rolled?

b) How many sides does each die have?

c) What is the probability that the four numbers rolled will **all be the same**?

d) What is the probability that the four numbers rolled will be **distinct**? (four different numbers, no repetitions)

e) What is the probability that at least one 1 and at least one 2 are rolled?

4 The Sieben Siegel (Seven Seals) card game uses a non-standard deck. The deck has **fifteen numbers** (1 – 15) in each of **five colors** (red, green, blue, yellow, and purple).

a) How many cards are in the deck?

b) If you are dealt six cards, what is the probability that they are **all the same color**?

c) If you are dealt six cards, what is the probabiilty that you have **two trios**? (3 of one kind and 3 of another kind)

d) If you are dealt six cards, what is the probabilty that you have **3-of-a-kind**? (3-of-a-kind means that 3 cards are the same number, and the other three are different from each other and from the three that match.)

Go back and make sure you didn't slip back into standard deck mode anywhere.

5 Consider the following game. A player rolls two **fair four-sided** dice and wins the difference between the two values. Here are a few examples:

roll	W
$(2,2)$	0
$(3,2)$	1
$(2,4)$	2

Let the random variable W be the amount won in a single play of the game.

a) Here is the probability table for W.

Value of W	0	1	2	3
Probability	1/4	3/8	1/4	1/8

Explain why $p(W=3)=1/8$.

b) What is $p(W>1)$?

c) What is the expected value of W?

d) What does the expected value of W tell us about the game? (That is, how do you interpret the number you just calculated?)

6 Calvin has three hats. Each hat contains some maroon and some gold marbles.

- Hat A has 1 maroon and 3 gold.
- Hat B has 1 maroon and 1 gold.
- Hat C has 9 maroon and 1 gold.

Calvin randomly selects a hat (with equal probability) and then randomly selects a marble from that hat.

a) Alltogether, there are 16 marbles. Explain why they are not equally likely to be selected.

b) What is the probability that the marble selected comes from hat A?

c) What is the probability that the selected marble is the maroon marble in hat A?

d) What is the probability that the selected marble is maroon (from any hat)?

e) If the marble selected is maroon, what is the probability that the hat selected is hat A?

f) Let M be the event that the marble selected is marroon. Let A be the event that the hat selected is hat A. Circle the correct notation for the probabilities in parts b – e above.

part b: $p(A)$ $p(M)$ $p(A \cap M)$ $p(A \cup M)$ $p(A \mid M)$ $p(M \mid A)$

part c: $p(A)$ $p(M)$ $p(A \cap M)$ $p(A \cup M)$ $p(A \mid M)$ $p(M \mid A)$

part d: $p(A)$ $p(M)$ $p(A \cap M)$ $p(A \cup M)$ $p(A \mid M)$ $p(M \mid A)$

part e: $p(A)$ $p(M)$ $p(A \cap M)$ $p(A \cup M)$ $p(A \mid M)$ $p(M \mid A)$

Space for additional work.

Solutions Problems

1

```
BasicProb

## # A tibble: 5 x 3
##   probAUB probAgivenB probBgivenA
##     <dbl>       <dbl>       <dbl>
## 1     0.8       0.25        0.2
## 2     0.7       0.750       0.5
## 3     0.6       0.5         0.2
## 4     0.7       0.667       0.333
## 5     0.6       0.6         0.750
```

2

```
dplyr::tibble(
  n_dice = n_dice,
  n_six  = n_six,
  prob   = dbinom(n_six, n_dice, 1/6),
  prob2  = choose(n_dice, n_six) * (1/6)^n_six * (5/6)^(n_dice - n_six))

## # A tibble: 5 x 4
##   n_dice n_six   prob  prob2
##    <dbl> <dbl>  <dbl>  <dbl>
## 1     10     3 0.155  0.155
## 2     11     4 0.0711 0.0711
## 3     12     4 0.0888 0.0888
## 4     13     3 0.214  0.214
## 5     14     4 0.125  0.125
```

3

```
# all the same
4 / 4^4

## [1] 0.015625

# all distinct
4*3*2*1 / 4^4

## [1] 0.09375

# P(A) = P(at least one 1) = 1 - P(no 1's)
1 - (3/4)^4

## [1] 0.6835938

# P(A and B) = P(A) + P(B) - P(A union B)
# = P(at least 1 1) + P(at least 1 2) - (1  - P(no 1's and no 2's))
(1 - (3/4)^4) * 2 - (1 - (1/2)^4)

## [1] 0.4296875

(1 - (3/4)^4) * 2 - (1 - (1/2)^4) %>% fractions

## [1] 55/128
```

4

```r
15 * 5

## [1] 75

# 3 pairs
choose(15,3) * choose(5,2) * choose(5,2) * choose(5,2) / choose(75, 6)

## [1] 0.00225964

# 2 trios
choose(15,2) * choose(5,3) * choose(5,3) / choose(75, 6)

## [1] 0.00005214553

# 4-of-a-kind
choose(15,1) * choose(5,4) * choose(14,2) * choose(5,1) * choose(5,1) / choose(75, 6)

## [1] 0.0008473648

# 3-of-a-kind
choose(15,1) * choose(5,3) * choose(14,3) * choose(5,1) * choose(5,1) * choose(5,1) / choose(75, 6)

## [1] 0.03389459
```

Unused Problems

7 Let $X \sim \mathsf{Binom}(12, 0.75)$. What is $p(X = 10)$?

8 Dice.

a) If you roll three standard dice (6 sides, each with equal probability), what is the probability that all three numbers will be the same?

b) If you roll three standard dice, what is the probability that the three numbers rolled will be distinct (4 different numbers, no repetitions)?

c) If you roll three standard dice, what is the probability that **at least one** of the numbers rolled is a **six**?

d) If you roll three standard dice, what is the probability that **exactly one** of the numbers rolled is a **six**?

9 One of the variations of the the Michigan Lottery Daily 4 game has two possible prizes. The big prize is \$3200. The probability of winning this prize is 1/10000. The smaller prize is \$360. The probability of winning this prize is 1/2500.[1] Of course, if you don't win one of those prizes, you get nothing.

Let W be the amount of money won from one of these lottery tickets.

a) What is $p(W = 0)$?

b) What is the expected value of W?

c) What does the expected value of W tell us about this lottery game. (Note: It costs \$1 to buy one of these tickets).

d) What is the variance of W?

10 The Five Crowns card game uses a non-standard deck. The deck has **five suits** (clubs, diamonds, hearts, spades, and stars). Each suit has **11 kinds** (3 through King, there are no aces and no 2's). There are also jokers, but for this problem they have been removed. That leaves a deck with **55 cards**.

a) If you are dealt six cards, what is the probability that they are all the same suit?

b) If you are dealt six cards, what is the probabiilty that you have two trios? (3 of one kind and 3 of another kind)

c) If you are dealt six cards, what is the probabilty that you have 3-of-a-kind? (3-of-a-kind means that 3 cards are the same kind, and the other three are different from each other and from the 3 that match.)

Go back and make sure you didn't slip back into standard deck mode anywhere.

11 A fair coin is flipped 6 times.

a) What is the probability that there are exactly 4 heads?

b) What is the probability that there are at least 4 heads?

c) What is the probability that there are at least 4 *consecutive* heads?

[1] This information is available on their website in a different form – and has some errors because they mix up odds and probability. I've fixed those errors here.

12 ABC-123 LOTTERY. Consider the following lottery type game: A player selects (in order) three *distinct* letters from A to Z and three *distinct* digits from 0 to 9. (Examples: one lottery ticket would be AZD-924, another would be ADZ-492. On the other hand, ABA-012 is not a legal ticket because A is repeated and 123-ABC is not a legal ticket because the numbers come first.) Later, the lottery commission draws three distinct letters and three distinct digits at random and in order.

- If all the numbers and letters match in order, then the player wins \$1,000,000.
- If all the numbers and letters match but are not in the correct order, then the player wins \$100,000.
- If the all the letters match (not necessarily in order), but the numbers do not, then the player wins \$1,000;
- If the all the numbers match (not necessarily in order), but the letters do not, then the player wins a prize of \$10;

a) Fill in the following table:

amount won	1,000,000	100,000	1000	10	0
probability					

b) Determine the **expected value** of the amount won from one lottery ticket. [This will be graded based on your answers to the questions above, even if they are incorrect, so be sure to show your work.]

c) How much must the lottery charge per play to avoid losing money at this game?

d) For which kind of prize should the lottery commission expect to pay out the most money? Explain.

13 Binomial coeffcients.

a) Show how to calculate $C(10,3) = \binom{10}{3} =$ **without a calculator. Show your work.**

b) Write a counting problem for which $\binom{10}{3}$ gives the anwer. (Keep your problem simple.)

14 This problem is about permutations of the 6-element set $A = \{1,2,3,4,5,6\}$

a) Put the following permutations in lexicographical order: 135246, 124536, 145326, 145362.

b) List the next **FOUR** permutations in lexicographical order after 124653.

15 Sums.

a) Express the following sum using $\sum$ notation:

$$5 + 7 + 9 + 11 + 13 + \cdots 5005$$

b) Express the following sum using $\sum$-notation:

$$s(n) = 3 + 6 + 12 + \cdots + 3 \cdot 2^n$$

c) Express the value of $s(n)$ without using $\cdots$ or $\sum$-notation.

16 A standard deck of cards has 52 cards (4 suits of 13 kinds each). In 5-card poker, two pairs is 2 of one kind of card, 2 of another kind, and 1 of a third kind (so that it doesn't become a full house). (e.g. 5$\heartsuit$, 5$\clubsuit$, 8$\diamondsuit$, 8$\spadesuit$, 2$\spadesuit$) What is the probability of being dealt 2 pairs?

17 A bagel shop has 6 types of bagels. How many ways are there to select a ten bagels?